



# CAN YILMAZ

BİLGİSAYAR MÜHENDİSİ | COMPUTER ENGINEER



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## EDUCATION

**ONDOKUZ MAYIS  
UNIVERSITY**

Computer Engineer

2020-2025

GPA: 2.89/4

## LANGUAGES

- English *B2 level*
- Japanese *A1 level*
- German *A1 level*

## SKILLS

### PROGRAMMING LANGUAGES

- Python
- C++
- C#
- Java
- Ruby

### DATABASES

- Microsoft SQL Server

### TOOLS & TECHNOLOGIES

- Docker
- Unity
- MS Office
- Wireshark
- Contiki
- Scikit-learn
- Zemberek NLP
- BeautifulSoup
- Tkinter
- JSBSim
- JADX-GUI
- Android Studio
- PyQt5, 6
- FlightGear
- MissionPlanner
- MAVLink
- OpenCV
- PX4 + ArduPilot
- Gazebo
- YOLO

## ABOUT ME

Hello, I'm Can Yılmaz, a 25-year-old Computer Engineer. I completed my primary and middle school education at Doğa College on a scholarship and graduated from Final Schools for high school.

I have been interested in computers and video games since childhood. This interest has not only shaped my passion for technology but has also contributed significantly to my personal and language development. Driven by this interest, I studied Computer Engineering at Ondokuz Mayıs University from 2020 to 2025.

I am curious and eager to learn and explore new things. I value continuous improvement and teamwork, and I am looking for a full-time position in an organization where I can develop my skills, keep up with new technologies, and contribute to shaping them.

## WORK EXPERIENCES

### Company Name

DROHYTECH Teknoloji ~ 16.05.2025 - Present

### Position

Software Development Engineer

### Description

As a software engineer working on fixed-wing unmanned aerial vehicles (UAVs), I work across flight control, simulation, and Ground Control Station (GCS) development.

- Developed projects integrating computer vision, machine learning, autonomous target engagement algorithms, and autonomous decision-making systems into flight software for real-time mission execution.
- Worked with FlightGear, JSBSim, ArduPilot, PX4, and Gazebo-based simulations, integrating aerodynamic and sensor testing with automated systems.
- Designed a Flight Termination Algorithm (FTA) as part of flight safety and mission assurance.
- Continue to develop professional GCS solutions using hybrid C++ / Python / QML architectures for defense industry applications.

### Company Name

BITES Defence & Aerospace ~ 31.07.2023 - 26.09.2023

### Position

Intern

### Description

Developed a Primary Flight Display (PFD) application in C# as part of my internship project.

## EXPERIENCES (DROHYTECH)

### Autonomous Target Guidance and Guidance Algorithms

- Developed a control system for a fixed-wing UAV in a FlightGear + ArduPilot SITL + JSBSim simulation environment, enabling autonomous target detection, guidance, and precision approach maneuvers. Designed a Python/pymavlink-based Finite State Machine (FSM) consisting of CRUISE → ALIGN → ROLL\_OUT → DIVE phases.
- Implemented continuous Line-of-Sight (LOS) angle tracking and Proportional Navigation (ProNav) guidance algorithms. Added target velocity estimation and lead-time prediction to enable dynamic timing based on closing velocity.
- Resolved PID controller oscillation issues through low-pass rate filtering and phase-based gain tuning, improving fixed-target approach accuracy from approximately 300–900 m to 13–30 m.
- Reworked the telemetry data pipeline from an initial Telnet-based synchronous approach (~1 Hz) to FlightGear's native UDP protocol, achieving approximately 20 Hz real-time communication with lower latency.
- Defined missing aerodynamic stability derivatives (Clp, Cm<sub>q</sub>, Cnr, C<sub>m</sub>alpha, Clbeta, Cnbeta) in the JSBSim flight dynamics model to improve the aircraft's natural stability. Identified and addressed root causes including Euler-angle singularities, rudder-roll aerodynamic coupling, and LOS calculation singularities using feed-forward compensation and gradual gain transitions.
- Created a 3D target model in Blender and integrated it into FlightGear's AI scenario system. Optimized ArduPilot SITL parameters and engine/frame configurations, including the plane-jet frame and AUTOTUNE settings, based on flight performance requirements.



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## • PX4 & Gazebo Integration (System and Sensor Simulation)

- Integrated a PX4-based flight control system with Gazebo for hardware and sensor simulation.
- Conducted system tests using simulated IMU, GPS, barometer, and LiDAR sensors.
- Integrated a camera into the simulation environment and performed OpenCV-based image processing and YOLO object detection on the video stream

## Ground Control Station (GCS)

- Developing a professional, modular GCS software based on Qt/QML and C++, using a hybrid C++ / Python / QML architecture.
- Built a multi-module interface including Dashboard, Map, Mission Planning, Video, Logs, Commands, and Parameters. Developed a Parameter Panel capable of managing 1,000+ autopilot parameters with search, listing, and live value updates.
- Implemented real-time bidirectional telemetry and control over MAVLink with UDP and serial connection support.
- Integrated flight mode switching (MANUAL, AUTO, CRUISE, GUIDED, RTL, etc.), ARM/DISARM, TAKEOFF/LAND, and emergency functions such as Kill Switch and Emergency Land into a unified control interface.
- Developed an RC Override panel with live sliders for manual Pitch, Roll, Yaw, and Throttle control.
- Designed a draggable and dockable widget-based interface on an OpenStreetMap-based map, including live vehicle position, Attitude Indicator, ILS/Localizer data, and Mission List.
- Integrated real-time video streaming using GStreamer, UDP/RTP, and H.264, along with YOLO-based AI object tracking for real-time analysis of UAV imagery.
- Added support for analyzing recorded video and reviewing object tracking over historical flight data.
- Implemented a command-line interface for sending free-text commands such as arm, disarm, mode auto, and takeoff for rapid interaction with the vehicle.

## Computer Vision & AI

- Developed object detection and tracking systems using YOLOv5, YOLOv7, YOLOv8, and YOLOv11.
- Managed the complete machine learning workflow, including dataset collection, annotation, model training, and testing.
- Performed image processing, masking, and real-time object tracking using OpenCV.
- Integrated trained models into a MAVLink-based GCS software and used them for real-time target recognition and autonomous decision-making.
- Designed AI modules to support autonomous mission changes and target prioritization during flight.

## Flight Simulation & Aerodynamic Testing

- Conducted HIL/SIL tests on a fixed-wing UAV model using ArduPilot, FlightGear, and JSBSim.
- Analyzed aerodynamic behavior and integrated the results with autonomous control systems.
- Designed a Flight Termination Algorithm (FTA) for autonomous mission scenarios, providing a safe flight termination mechanism in the event of control loss or other risk conditions.
- Managed MAVLink-based data exchange between the GCS and flight controller to evaluate FTA performance in the simulation environment.

## EXPERIENCES (OTHER)

### Software Engineering Intern – BITES Defence & Aerospace

- Developed a Primary Flight Display (PFD) simulation application using C#.
- Designed a dynamic interface for collecting altitude, temperature, heading, airspeed, aircraft model/type, and weather data, and stored the data in text-file format.
- Developed an interface and data transmission module for sending data over UDP.
- Designed the main application to process incoming data and display it on the PFD interface in real time.
- Created test messages and scenarios to validate incoming and transmitted data.
- Developed a data validation system that generated HTML reports, visually highlighting valid data in green and invalid data in red.
- Analyzed network traffic using Wireshark to verify the correct transmission of UDP packets and support protocol-level debugging.
- Prepared system requirements and technical documentation for the project.
- Worked with real-time data processing and communication systems, focusing on the management and validation of flight data in a simulation environment.
- Designed test scenarios to simulate error conditions and evaluate system robustness.



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## Projects & Technical Experience (Ondokuz Mayıs University)

### Derin Öğrenme Tabanlı Nesne Tespiti – Kalem ve Telefon Sınıflandırması (Real-Time Object Detection)

#### Dataset Structure and Annotation:

- Created a custom dataset consisting of three categories: pen, phone, and negative samples.
- Annotated the images using CVAT and exported the annotations in YOLO format.
- Captured screenshots through the Windows API and automatically classified the collected images.

#### Model Training and Usage:

- Trained a custom YOLOv8 object detection model for two target classes: pen and phone.
- Monitored data augmentation, training/validation splits, mean Average Precision (mAP), and loss metrics throughout the training process.
- Trained the model to detect and classify pens and phones with high accuracy.

#### Real-Time Application Development:

- Developed a real-time object detection system using OpenCV and a webcam.
- Displayed the detected class name and confidence percentage along with a bounding box when a pen or phone was placed in front of the camera.

#### Performance and Optimization:

- Used a confidence threshold to control the reliability of model predictions.
- Displayed a "Not confident" message when the prediction confidence was below the defined threshold.
- Also experimented with the TensorFlow-based MobileNetV2 architecture, but selected YOLOv8 due to its more efficient performance in the project.

### Drone Station Assignment Optimization – (Three-Person Graduation Project)

#### Project Objective:

- Developed an energy optimization approach for assigning drones to suitable stations, aiming to find efficient solutions using PSO, GWO, ACO, Genetic Algorithm, ABC, Grasshopper Optimization Algorithm, and Differential Evolution.

#### Algorithm Research and Selection:

- Investigated seven different optimization algorithms through a literature review: Particle Swarm Optimization (PSO), Grey Wolf Optimization (GWO), Ant Colony Optimization (ACO), Genetic Algorithm (GA), Artificial Bee Colony (ABC), Grasshopper Optimization Algorithm (GOA), and Differential Evolution (DE).
- Based on the experimental results, focused on PSO and GWO, which provided the most suitable solutions for the defined scenarios.

#### Optimization Process:

- Implemented PSO and GWO in Python for drone station assignment and energy optimization.
- Optimized algorithm parameters according to different drone and station scenarios and evaluated the resulting solutions.
- Performed optimization over 10,000 iterations and visualized the results.

#### Technologies & Tools:

- Python, Tkinter (GUI), Matplotlib, PSO, GWO

### Wireless Network Simulation & Encryption

- Conducted wireless network simulations using Docker, Contiki-NG, and Cooja.
- Applied and analyzed different encryption techniques for secure data transmission in simulated network environments.
- Analyzed network traffic using Wireshark and examined packet flows in detail.

### Reverse Engineering & Security Analysis

- Conducted basic reverse engineering studies using Java.
- Analyzed the code structures of various software applications and investigated potential security vulnerabilities.

### Dental Clinic Management System - C#

- Developed a desktop dental clinic management system using C#.
- Designed and implemented core features including patient records, appointment tracking, billing, and doctor scheduling.
- Used SQL Server for database management and secure storage and processing of patient information.

### Graph Algorithms & Sorting Application

- Developed an application implementing different graph and sorting algorithms.
- Practiced data structures and algorithms while analyzing their performance.

### Customer Retail Database Application — (Three-Person Team Project)

- Developed a customer retail database application as part of a three-person team.
- Took an active role in database design, query optimization, and user interface development.
- Implemented core functionality for customer records, inventory tracking, and sales analysis.
- Performed optimization work to improve system performance and data security.

### Machine Learning & Artificial Neural Networks

- Developed an Artificial Neural Network-based multiplication prediction model using Python and Scikit-Learn's MLPRegressor.
- Split the dataset into training and test sets and evaluated model performance using Mean Squared Error (MSE).
- Applied data normalization to improve prediction stability.
- Tested the model with sample inputs and analyzed the resulting predictions.